

### 8.4.3 Single-Junction Form Loss Data, CCC0111

This card is optional. The user-specified form loss is given in Words 4 and 5 of card CCC0101 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where  $K_F$  and  $K_R$  are the forward and reverse form loss coefficient.  $A_F$  and  $A_R$  are the Words 4 and 5 of card CCC0101.  $Re$  is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for  $A_F$  and  $A_R$ ; the same information applies for  $B_F$  and  $B_R$ .

W1(R)  $B_F (\geq 0)$ . This quantity must be greater than or equal to 0.

W2(R)  $C_F (\geq 0)$ . This quantity must be greater than or equal to 0.

W3(R)  $B_R (\geq 0)$ . This quantity must be greater than or equal to 0.

W4(R)  $C_R (\geq 0)$ . This quantity must be greater than or equal to 0.

### 8.4.4 Single-Junction Initial Conditions, CCC0201

This card is required for single-junction components.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flows.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) [ $W1 = 0$ ] or mass flow (kg/s, lb/s) [ $W1 = 1$ ].

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) [ $W1 = 0$ ] or mass flow (kg/s, lb/s) [ $W1 = 1$ ].

W4(R) Interface velocity (m/s, ft/s). Enter 0.

## 8.5 Time-Dependent Junction Component, TMDPJUN

This component is indicated by TMDPJUN for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the time-dependent junction component is numbered as CCC000000.

### 8.5.1 Time-Dependent Junction Geometry, CCC0101

This card is required for time-dependent junction components.

- W1(I)      From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. A nonzero N specifies the expanded format. The number N equal to 1 and 2 specifies the inlet and outlet faces, respectively, for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.
- W2(I)      To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.
- W3(R)      Area ( $\text{m}^2$ ,  $\text{ft}^2$ ). If 0, the area is set to the minimum flow area of the adjoining volumes. There are no junction area restrictions for time-dependent junctions.
- W4(I)      Junction control flag. This word is optional and has the format jefvcahs. It is not necessary to input leading zeros. Only the e-flag can be set. The other flags have to be 0, so either 0 or 1000000 are the only two allowable entries.
- e-flag      The digit e specifies the modified PV term in the energy equations. e = 0 means that the modified PV term will not be applied, and e = 1 means that it will be applied.

### 8.5.2 Time-Dependent Junction Data Control Word, CCC0200

This card is optional. If this card is missing, the second and third words of the time-dependent data are assumed to be velocities.

- W1(I)      Control word. If 0, the second and third words of the time-dependent junction data in cards CCC0201-0299 are velocities. If 1, the second and third words of the time-dependent junction data in cards CCC0201-0299 are mass flows. In both cases, the fourth word is interface velocity and should be entered as 0.
- W2(I)      Table trip number. This word is optional. If missing or 0 and Word 3 is missing, no trip is used, and the time argument is the advancement time. If nonzero and Word 3 is missing, this number is the trip number and the time argument is -1.0 if the trip is false, and the advancement time minus the trip time if the trip is true.

- W3(A)      Alphanumeric part of variable request code. This quantity is optional. If present, this word and the next are a variable request code that specifies the search argument for the table lookup and interpolation. If the trip number is 0, the specified argument is always used. If the trip number is nonzero, -1.0E75 is used if the trip is false, and the specified argument is used if the trip is true. TIME can be selected, but note that the trip logic is different than if this word is omitted. The variable MFLOWJ should not be used as a search variable; a user-initialized control variable that uses MFLOWJ should be used instead.
- W4(I)      Numeric part of variable request code. This is assumed 0 if missing.

### 8.5.3 Time-Dependent Junction Data, CCC0201-0299

These cards are required for time-dependent junction components. A set of data consists of the search variable (e.g., time) followed by the required data indicated by control Word 1 on card CCC0200. The card numbers need not be consecutive, but the value of the search variable in a succeeding set must be equal to or greater than the value in the previous set. One or more sets of data up to 5000 sets may be entered. If the phase or material is not present, 0 may be entered for a velocity or flow. The interpolation and card formats for the time-dependent data are identical to that in Section 8.3.3 (cards CCC0201-0209, Time-Dependent Volume Data cards).

When doing a single-phase problem and entering velocities here, the same value should be entered for both liquid and vapor velocities. If entering mass flows, the correct value should be entered for either liquid or vapor (whichever single-phase is being modeled) and the other entry should be 0.

If the user wants to specify the vapor void fraction as a function of time in the time-dependent volume, and the total mass flow as a function of time in the time-dependent junction, then both the phasic (gas and liquid) mass flow rates must be calculated and entered in these cards.

- W1(R)      Search variable (e.g., time).
- W2(R)      Liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on control Word 1 on card CCC0200.
- W3(R)      Vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on control Word 1 on card CCC0200.
- W4(R)      Interface velocity (m/s, ft/s). This feature has not been implemented yet, so enter 0.

As described above, sets may be entered one or more per card.